IAS PRELIMS/MAINS

SCIENCE & TECHNOLOGY

A COMPREHENSIVE COVERAGE

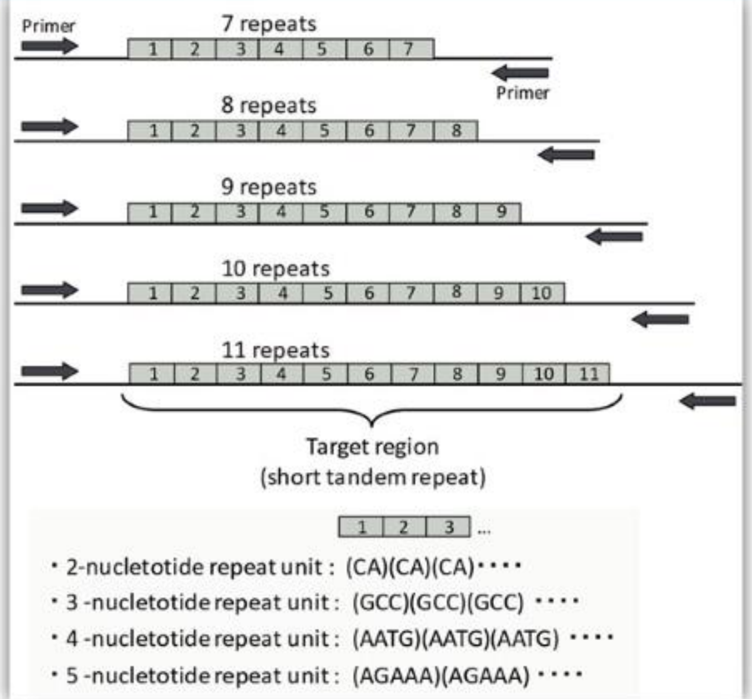
MODULE - 17

BIOTECHNOLOGY - 7

Learning Space

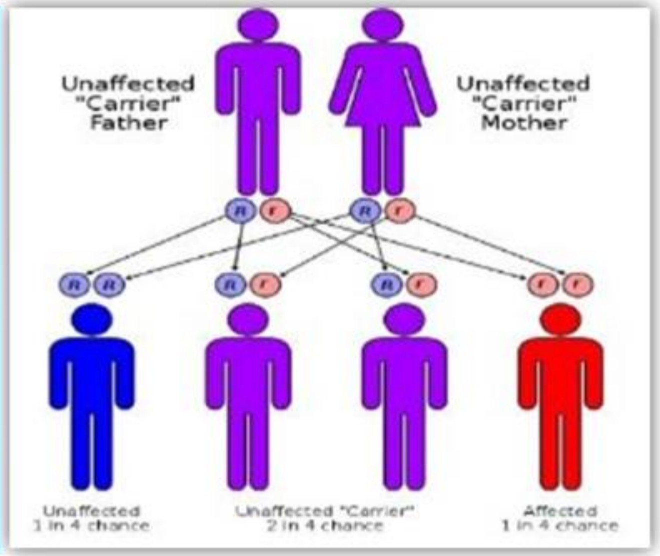
DNA FINGER PRINTING

- Also called DNA typing and DNA profiling.
- Every individual human has unique finger prints. These have been used for identification since a long time, but these can be altered by surgery.
- A sequence of nucleotides in DNA molecule is also unique for a person and that unique information is used in DNA finger printing. The unique sequence is same for every cell and cannot be altered by any known method or treatment.
- Around 99.9% of all human genomes are similar but among the 0.1% (3 million pair of nucleotides) there are certain pairs of nucleotides that repeat themselves in the genome. These are called Short Tandem Repeats (STR).
- The STR vary not only in their sequence but also in the number of repeats of these sequences.
- So, using this the DNA samples found at a crime scene can be used to identify the culprit from the list of suspects.



DNA FINGER PRINTING

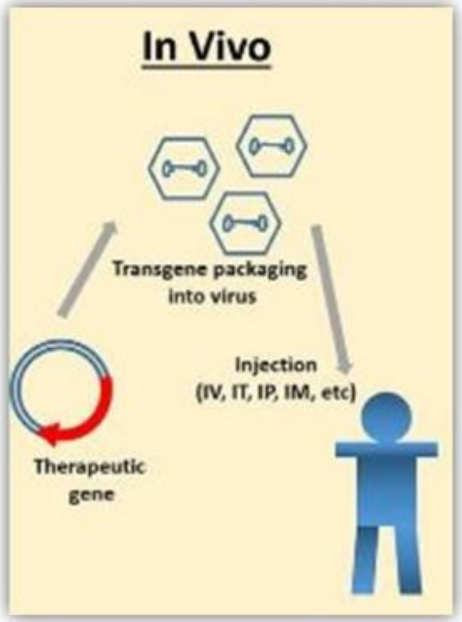
- DNA finger printing is used in
  1. Forensic labs for identification of criminals.
  2. To determine paternity.
  3. To identify dead bodies in any accident by comparing the DNA's of parents or children.
  4. To identify different racial groups and trace evolution.



Learning Space

GENE THERAPY

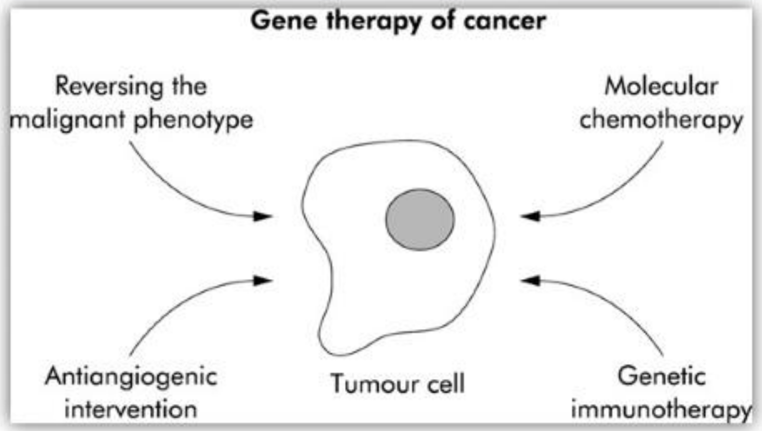
- Gene therapy is designed to introduce genetic material into cells to compensate for abnormal genes or to make an important protein.
- If a mutated gene causes a necessary protein to be faulty or missing, gene therapy may be able to introduce a normal copy of the gene to restore the protein.
- A gene that is inserted directly into a cell does not function. So, a vector is used to deliver the gene. Certain viruses are often used as vectors because they can deliver the gene by infecting the cell.
- The viruses are genetically modified so that they don't cause any disease when they are used as vectors.
- The vector can be given intravenously (IV) directly into a specific tissue in the body where it is taken up by individual cells (In Vivo Gene Therapy).
- Or, a sample of the patient's cells can be removed and exposed to the vector in a lab. The cells containing the vector are then returned to the patient.
  - ✓ Ex: Vivo Gene Therapy



GENE THERAPY

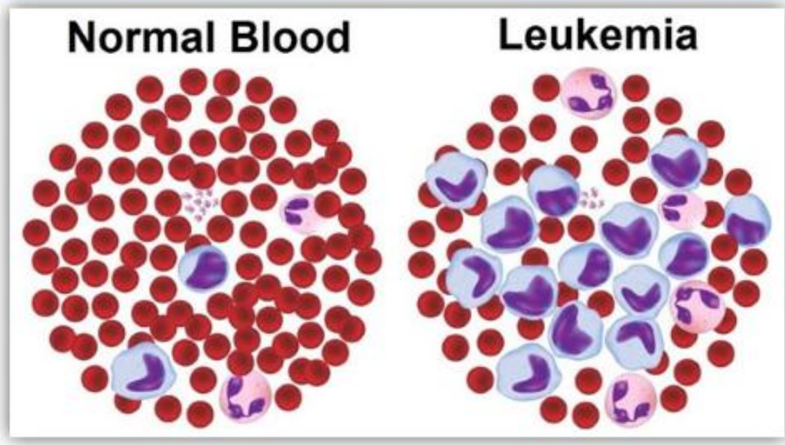
ADVANTAGES

1. Gene therapy can cure genetic diseases that have no other treatment.
2. Gene therapy can be used for cancer treatment, to kill the cancerous cells.
3. Gene expression can be controlled.



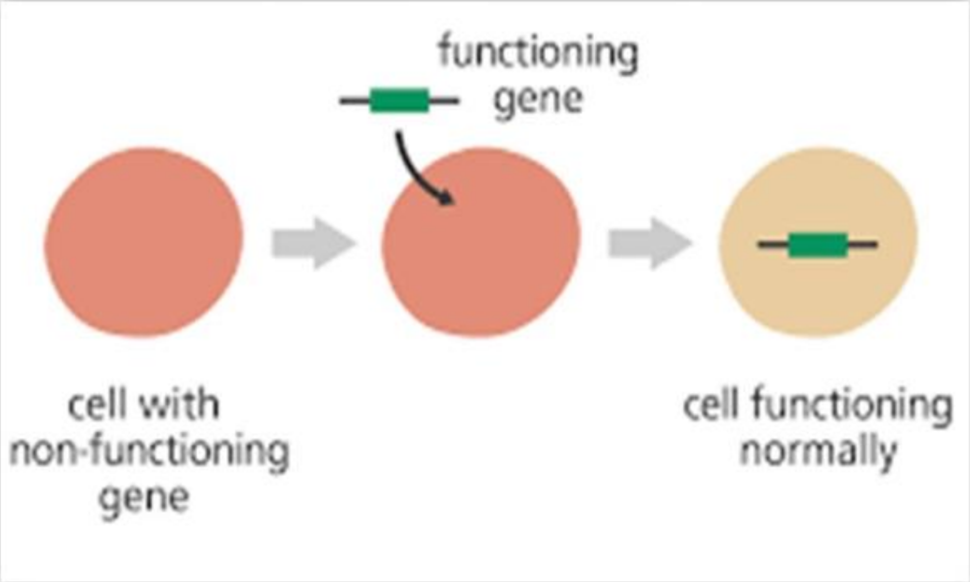
ISSUES

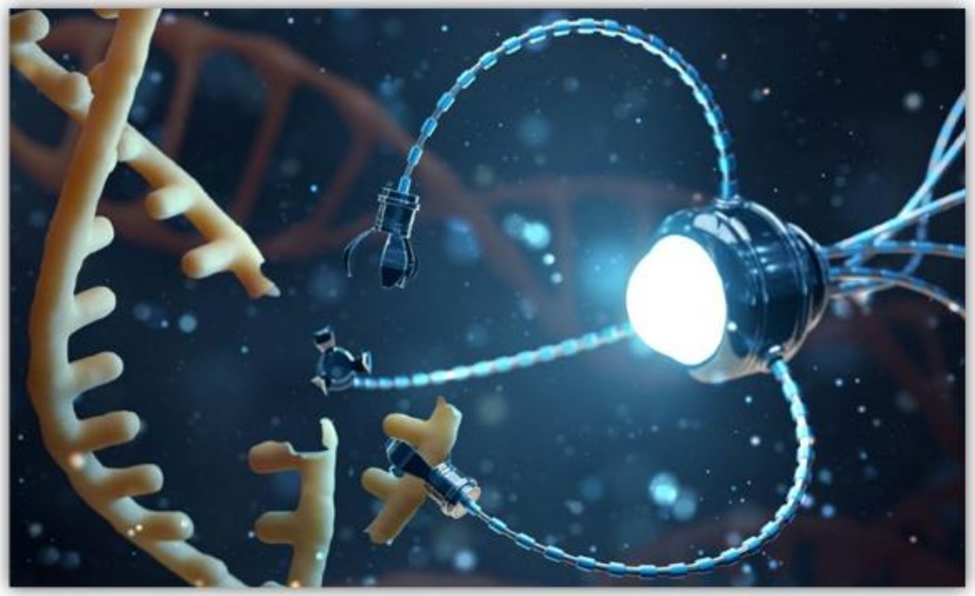
1. Immune response to viral vectors/foreign genes.
2. Newly introduced genes can disrupt important functioning genes in the target cells leading to cancers.
3. Viral vectors can cause cancers such as leukemia.
4. There is a risk of multiple organ failure due to gene therapy.



CURRENT STATUS OF GENE THERAPY

1. The technology is still in developmental stage and is yet to be developed to its fullest potential.
2. Many patients were treated by gene therapy so far mostly without long-term success.
3. The future success of gene therapy also depends on the advancements in other fields such as nanotechnology.





Learning Space

PHARMACOGENOMICS

1. It is the study of how genes affect a person's response to drugs. This field combines pharmacology (study of drugs) and genomics (the study of genes and their functions) to develop effective and safe medications and doses which are tailor made to a person's genetic makeup.
2. Many drugs that are currently available are 'one size fits all', but they don't work in the same way for everyone.
3. Without studying the genetic makeup, it is difficult to predict
   - ✓ who will benefit from a medication
   - ✓ who will not respond at all and
   - ✓ who will experience negative side effects.

Currently the field of Pharmacogenomics is still in infancy. In the future pharmacogenomics will help in the development of tailored drugs to treat a wide range of health problems.

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0866 244 44 72
98499 42299
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For Suggestions:
suggestions@learningspace.in

To Contact us:
info@learningspace.in

For Information:
info.learningspacedigital@gmail.com